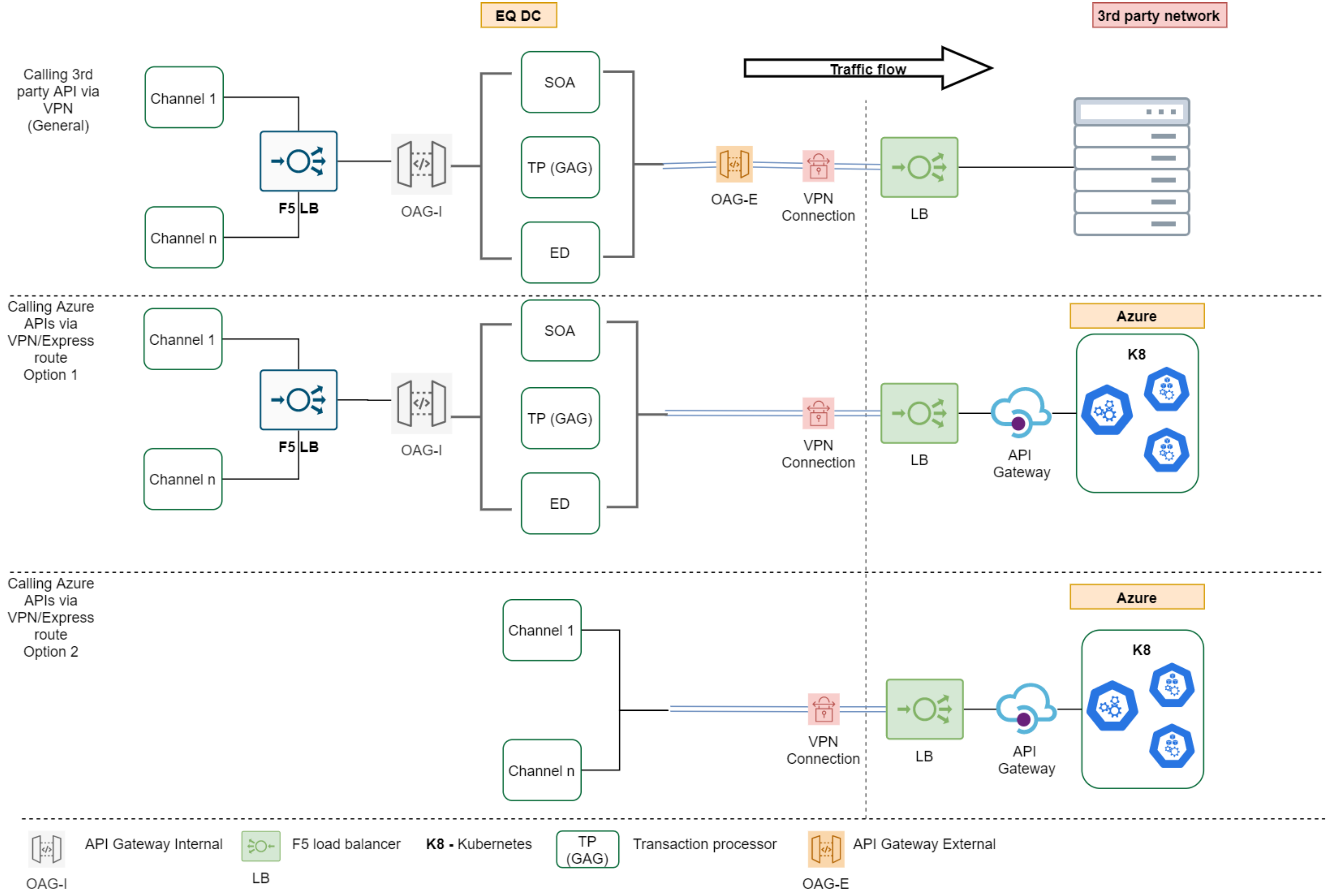
API Integration patterns

- Outbound APIs – VPN
- Outbound APIs - Internet
- Inbound APIs (To Equity network)
- Internal API calls

Outbound APIs - VPN



Architecture description (VPN) - Current

- Calls to external APIs must go through Internal OAG
- OAG will then call SOA or TP or ED depending on functionality required
- For transitioning purposes, SOA, TP and ED will call the external API via VPN. Target is to have only one ESB call external APIs through External OAG
- This will ensure single point of API entry & exit
- It will be easy to standardize, secure and govern services

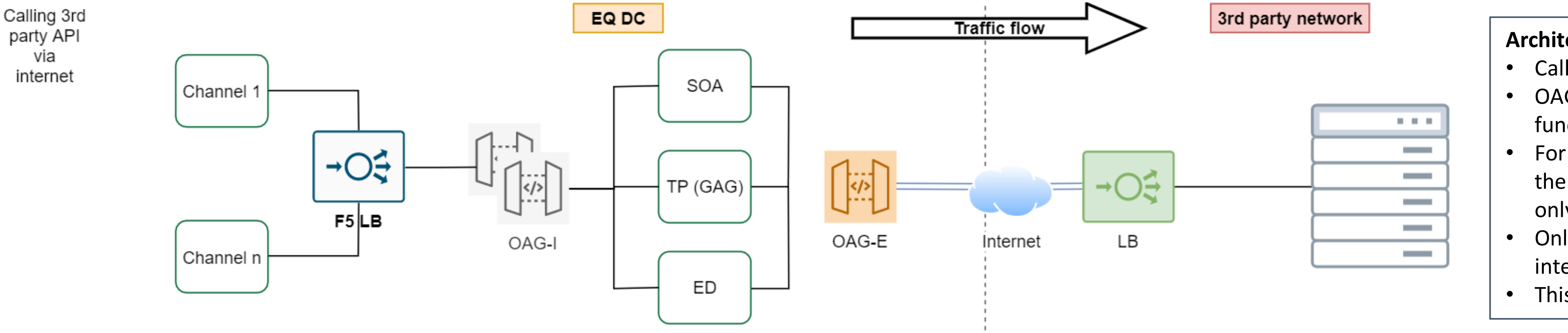
Architecture description (VPN-Azure).

- Azure will be considered as an Equity data centre
- Where we already have internal APIs consuming services on Azure, calls to Azure APIs by internal apps will go through the internal API Gateway
- This will be for cases where additional processing is required (e.g call to FI, message formatting etc)
- SOA, TP or ED will call Azure API gateway
- Entry to any API resource is via API gateway whether internal or on Azure

Architecture description (VPN-Azure).

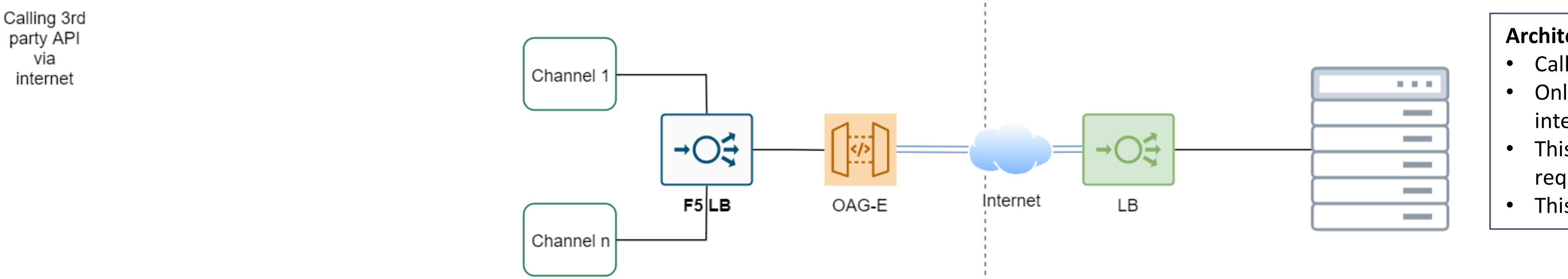
- Azure will be considered as an Equity data centre
- Calls to Azure APIs by internal apps will go to Azure API Gateway. Azure API gateway will have an internal IP accessible by internal systems
- Since Azure is an extension of our DC, we don't need to go through internal or external OAG

Outbound APIs - Internet



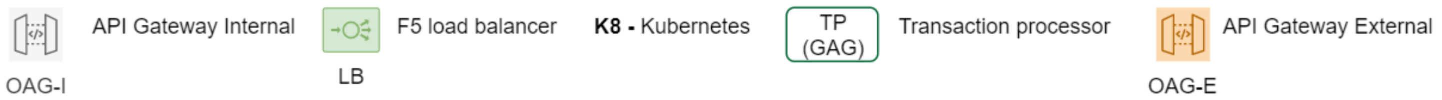
Architecture description (Option 1)

- Calls to external APIs will go through Internal OAG
- OAG will then call SOA or TP or ED depending on functionality required
- For transitioning purposes, SOA, TP and ED will call the external API via External OAG. Target is to have only one System call external OAG
- Only external OAG will be allowed to connect to internet APIs on the firewalls.
- This will ensure single point of API entry & exit



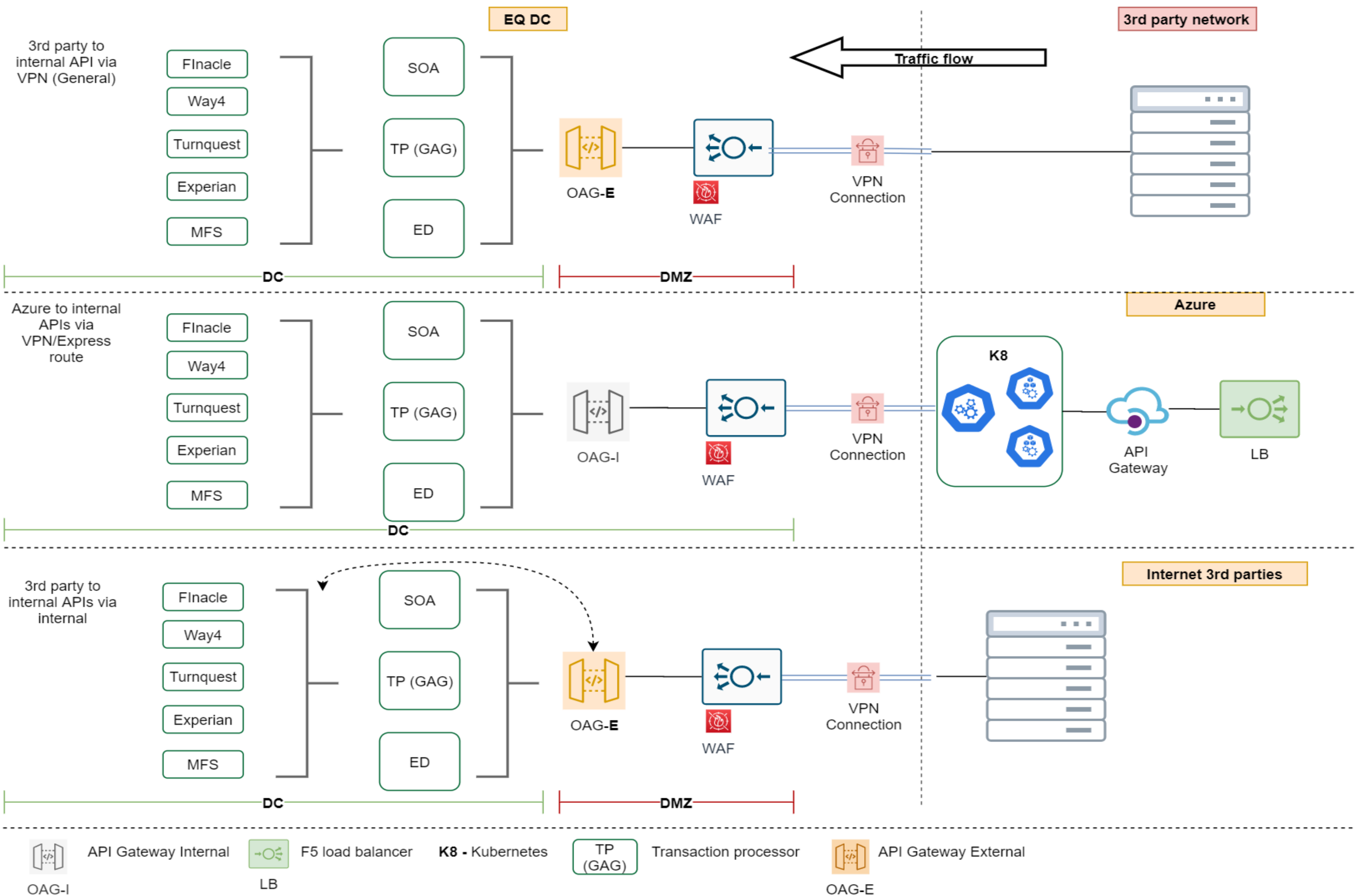
Architecture description (Option 2)

- Calls to external APIs will go through External OAG
- Only external OAG will be allowed to connect to internet APIs on the firewalls.
- This is for cases where no internal processing is required and ESB proxy service is unnecessary
- This will ensure single point of API exit



Inbound APIs

EQUITY



Architecture description (VPN)

- 3rd party calls to internal APIs will come through Internal OAG (Via LB)
- Internal OAG will then call SOA or TP or ED depending on functionality required
- SOA, ED and TP will fulfil the request on backend systems.

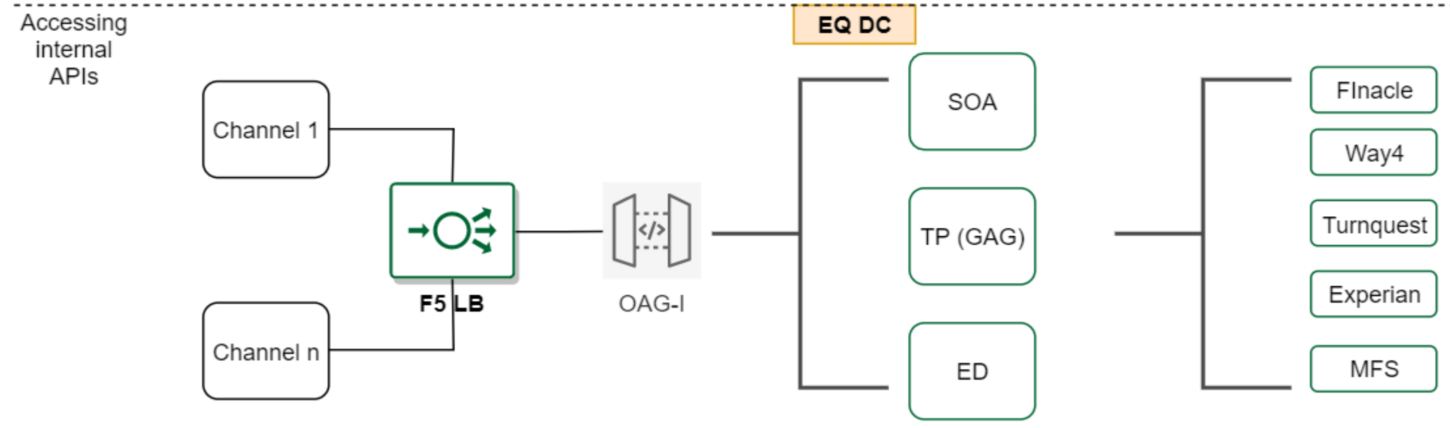
Architecture description (VPN-Azure)

- Calls from kubernetes services to internal APIs will come through Internal OAG (Via LB)
- Internal OAG will then call SOA or TP or ED depending on functionality required
- SOA, ED and TP will fulfil the request on backend systems.

Architecture description (Internet)

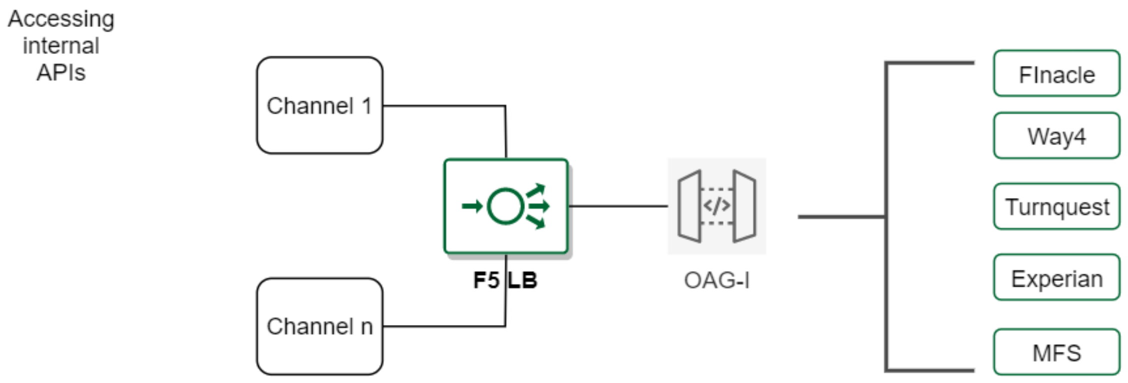
- Internet 3rd parties calls to internal APIs will come through External OAG (Via LB)
- External OAG will then call SOA or TP or ED depending on functionality required
- SOA, ED and TP will fulfil the request on backend systems.
- In other cases, external OAG will go directly to backend systems since some backed systems have standard APIs consumable from AOG

Internal APIs calls



Architecture description (Option 1)

- Channels, or any other application initiating an API call to another system must come through the internal OAG
- Internal OAG will then call SOA or TP or ED depending on functionality required
- SOA, ED and TP will fulfil the request on backend systems.
- In the future, only one ESB will connect to backend systems.



Architecture description (Option 2)

- Channels, or any other application initiating an API call to another system must come through the internal OAG
- Internal OAG will then calls backend system for fulfilment.
- This is for backend systems that have fairly standard APIs and OAG will require minimum adjustment to invoke them.